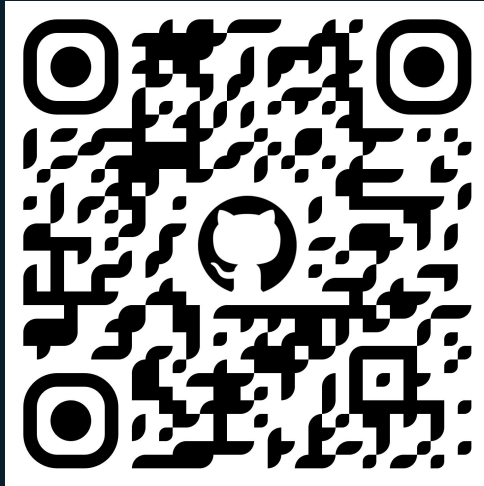


Resource



github.com/ro-witthawin/TBS-BI303-2026

TBS-BI303-2026



Class schedule

Date	Topic	Material
2026-08-03	Intro to Programming Language	Day01
2026-08-10	Basic String & List	Day02
2026-08-17	Function	Day03
2026-08-24	Loop	Day04
2026-08-31	Dictionary	Day05
2026-09-07	Class Object	Day06
2026-09-14	Freq Library	Day07
2026-09-21	Final Exam	

“List & Loop”

Basic List

[0, 1, 2, 3, 4]

Index

[0]

[1]

[2]

[3]

[4]

Negative
Index

[-5]

[-4]

[-3]

[-2]

[-1]

For-loop-in-List

```
for x_i in X:  
    print(x_i)
```

["a", 1, "b", 3, 4]



x_0



x_1



x_2



x_3



x_4

How to check <ITEM> in <LIST>

“a” in X

“c” not in X

X = [“a”, 1, “b”, 3, 4]

Empty List

`[]`

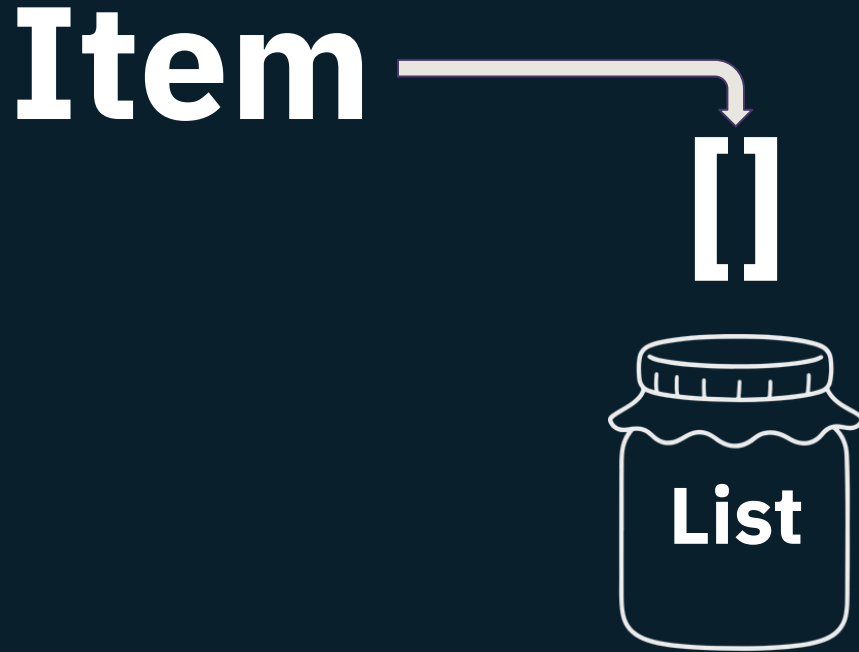


Empty List

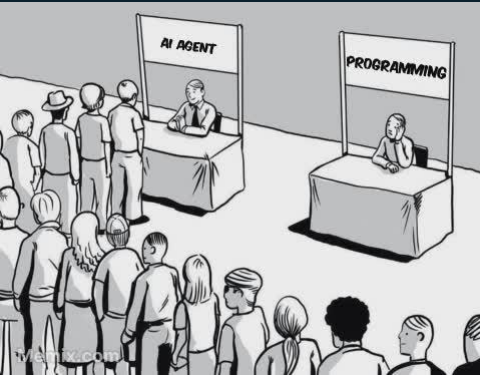
`[]` → **None**



List : append()



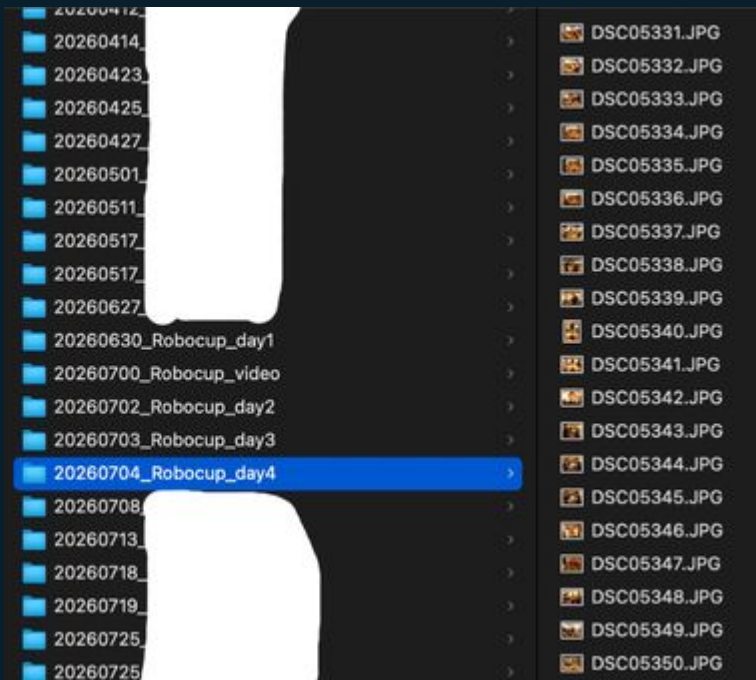
Problem :
we have many item in List
Hard to find something
Or Specific spec



“Dict”

{}

Dict : Dictionary



```
cam = {
```

```
    "20260630": "Robocup_day1",  
    "20260700": "Robocup_video",  
    "20260702": "Robocup_day2",  
    "20260703": "Robocup_day3",  
    "20260704": "Robocup_day4",
```

```
}
```

Dict : Element

String (Mostly) or Number (Few)

{ KEY : VALUE }

Any Object

String, Number, List, Dict, Object

Dict : Accual case

```
student_information = {  
    "id": "xxxxxxx",  
    "name": "Witthawin",  
    "age": 20,  
    "enrollmentStatus": False,  
    "homework": [  
        {  
            "exercise_id": "BI303_0001",  
            "score": 10  
        },  
        {  
            "exercise_id": "BI303_0002",  
            "score": 5  
        }  
    ]  
}
```

Dict : from key to value

```
student_information = {  
    "id": "xxxxxxxx",  
    "name": "Witthawin",  
    "age": 20,  
    "enrollmentStatus": False,  
    "homework": [  
        {  
            "exercise_id": "BI303_0001",  
            "score": 10  
        },  
        {  
            "exercise_id": "BI303_0002",  
            "score": 5  
        }  
    ]  
}
```

VALUE = DICT_NAME[KEY]

```
>>>student_information["name"]  
# Return "Witthawin"
```

```
>>>student_information["age"]  
# Return 20
```

```
>>>student_information["homework"][1]["score"]  
# Return 5
```

Dict : Update value from key

```
student_information = {  
    "id": "xxxxxxx",  
    "name": "Witthawin",  
    "age": 20,  
    "enrollmentStatus": False,  
    "homework": [  
        {  
            "exercise_id": "BI303_0001",  
            "score": 10  
        },  
        {  
            "exercise_id": "BI303_0002",  
            "score": 5  
        }  
    ]  
}
```

DICT_NAME[KEY] = VALUE

```
>>>student_information["age"] = 30  
# Return None, But update value in dict
```

```
>>>student_information["enrollmentStatus"] = True  
# Return None, But update value in dict
```

Empty Dict



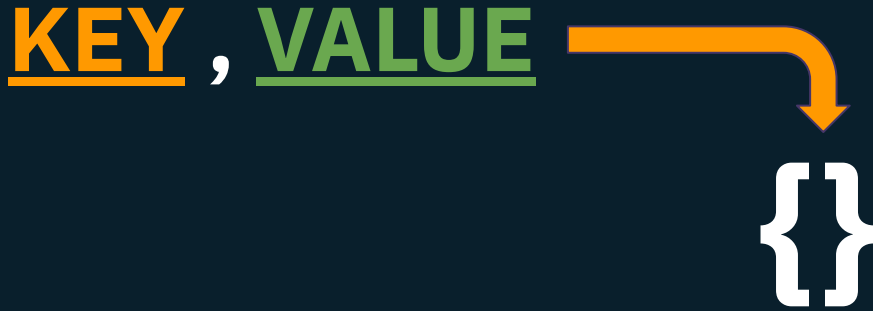
How to check key in Dict

“a” in X

“c” not in X

X = { . . . }

Add item in Dict



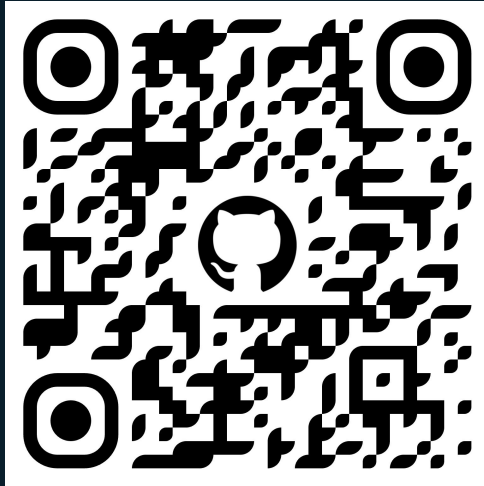
DICT_NAME[KEY] = VALUE

Dict : Accual case

```
student_information = {  
    "id": "xxxxxxx",  
    "name": "Witthawin",  
    "age": 20,  
    "enrollmentStatus": False,  
    "homework": [  
        {  
            "exercise_id": "BI303_0001",  
            "score": 10  
        },  
        {  
            "exercise_id": "BI303_0002",  
            "score": 5  
        }  
    ]  
}
```

```
{  
  "id": "07M0yfRBcAA",  
  "title": "เอกหักเพราะรักแม่เงิน",  
  "formats": [  
    {  
      "format_id": "sb3",  
      "format_note": "storyboard",  
      "ext": "mhtml",  
      "protocol": "mhtml",  
      "acodec": "none",  
      "vcodec": "none",  
      "url": "https://i.ytimg.com/sb/07M0yfRBcAA/storyboard3_L0/default.jpg?",  
      "width": 48,  
      "height": 27,  
      "fps": 0.00678334011667345,  
      "rows": 10,  
      "columns": 10,  
      "fragments": [  
        {  
          "url": "https://i.ytimg.com/sb/07M0yfRBcAA/storyboard3_L0/default.jpg?",  
          "duration": 14742.0  
        }  
      ],  
      "audio_ext": "none",  
      "video_ext": "none",  
      "vbr": 0,  
      "abr": 0,  
      "resolution": "48x27",  
      "aspect_ratio": 1.78,  
    }  
  ]  
}
```

Dict Operation in Colab



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